

Double Machine Learning

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1 Motivation

Many statistical methods have proven effective and efficient in many estimation problems. For example, we may use maximum likelihood estimation (MLE) to solve regression problems, and may use methods of moments (MoM), as well as generalized methods of moments (GMM), to solve problems based on moment conditions. When applying these methods, we may encounter the following problems.

Example 1.1 (Linear Regression with Controls). Suppose that we are interested in studying how years of education D affect annual income Y . To account for potential confounding effects, we control for several covariates that may also affect Y , including age x_1 , work experience x_2 , and region x_3 . We collect these covariates into a vector $\mathbf{x} = (x_1, x_2, x_3)^\top$.

Consider the following linear model:

$$Y = \theta D + \mathbf{x}^\top \boldsymbol{\beta} + \varepsilon, \quad \mathbb{E}(\varepsilon|D) = 0, \quad \mathbb{E}(\varepsilon|\mathbf{x}) = \mathbf{0}, \quad (1)$$

where θ is the effect of education on income and is the parameter of interest, $\boldsymbol{\beta}$ is the effect of the control variables, and ε denotes the error term.

The model implies the following moment condition:

$$\mathbb{E}[\psi(W; \theta, \boldsymbol{\beta})] \triangleq \mathbb{E}\left[\begin{pmatrix} D \\ \mathbf{x} \end{pmatrix} (Y - \theta D - \mathbf{x}^\top \boldsymbol{\beta})\right] = \mathbf{0}, \quad (2)$$

where $W = (Y, D, \mathbf{x})$ denotes the observed data, and $\psi(\cdot)$ is the corresponding moment function.

In practice, the sample size is typically large enough to estimate all parameters in this model.

Example 1.2 (Linear Regression with High-Dimensional Controls). Suppose that we are interested in estimating the causal effect of a drug treatment D on a patient's health outcome Y . To account for potential confounding effects, we collect genetic features, such as gene expression levels, as control variables. Let $\mathbf{x} \in \mathbb{R}^p$ denote the vector of genetic features.

In this example, we can similarly derive a linear model of the same form as Equation (1). However, since genetic features are high-dimensional, p can be much larger than the sample size n .

Example 1.3 (Partially Linear Model). Suppose that we are interested in studying the effect of a set of covariates $\mathbf{x}$ on an outcome Y , while allowing for a potentially nonlinear effect of another set of covariates $\mathbf{z}$. We consider the partially linear model

$$Y = \boldsymbol{\theta}^\top \mathbf{x} + g(\mathbf{z}) + \varepsilon, \quad (3)$$

where $\boldsymbol{\theta}$ is a finite-dimensional parameter of interest, $g(\cdot)$ is an unknown function, and ε is the noise satisfying

$$\mathbb{E}(\varepsilon \mid \mathbf{x}, \mathbf{z}) = 0. \quad (4)$$

Here, $g(\cdot)$ represents the potentially nonlinear effect of $\mathbf{z}$ on Y and serves as an infinite-dimensional nuisance parameter.

All three examples above have one thing in common: we must account for nuisance parameters in some problem settings, which can make parameter estimation more challenging.

In Example 1.1, the problem is not that severe. Since the nuisance parameter is low-dimensional ($p = 3$ in this example) and we can readily collect a sufficiently large amount of data, we can still obtain a good estimate of θ by minimizing the residuals or using methods of moments based on (2), despite the presence of nuisance parameters.

However, this problem can be quite arduous in Example 1.2. A classic way to estimate θ in this case is by LASSO or similar regularization methods, which can be formulated as

$$(\hat{\theta}, \hat{\boldsymbol{\beta}}) = \underset{\theta, \boldsymbol{\beta}}{\operatorname{argmin}} \mathbb{E} \left[(Y - \theta D - \mathbf{x}^\top \boldsymbol{\beta})^2 \right] + \lambda \|\boldsymbol{\beta}\|_1. \quad (5)$$

LASSO and other regularization methods inevitably introduce bias. If we fix $\hat{\boldsymbol{\beta}}$, the optimal θ can be derived directly by setting the partial derivative equal to 0:

$$\mathbb{E} \left[D \left(Y - \hat{\theta} D - \mathbf{x}^\top \hat{\boldsymbol{\beta}} \right) \right] = 0, \quad (6)$$

which yields

$$\begin{aligned} \hat{\theta} &= \frac{\mathbb{E}(DY) - \mathbb{E}(D\mathbf{x}^\top \hat{\boldsymbol{\beta}})}{\mathbb{E}(D^2)} \\ &= \frac{\mathbb{E} \left(D (\theta D + \mathbf{x}^\top \boldsymbol{\beta} + \varepsilon) \right) - \mathbb{E}(D\mathbf{x}^\top \hat{\boldsymbol{\beta}})}{\mathbb{E}(D^2)} \\ &= \theta + \mathbb{E} \left(D\mathbf{x}^\top (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) \right). \end{aligned} \quad (7)$$

Therefore, the bias can also affect the estimation of the parameter of interest. Fortunately, various debiasing methods have been developed to mitigate this issue, such as orthogonalization and bias correction.

The problem can be more challenging in Example 1.3, as the nuisance g is a function, which is infinitely dimensional. In this setting, $\boldsymbol{\theta}$ can be estimated via the partialling-out technique. We first take conditional expectation $\mathbb{E}(\cdot \mid \mathbf{z})$ on both sides:

$$\mathbb{E}(Y \mid \mathbf{z}) = \boldsymbol{\theta}^\top \mathbb{E}(\mathbf{x} \mid \mathbf{z}) + g(\mathbf{z}), \quad (8)$$

subtracting (8) from (3) yields

$$Y - \mathbb{E}(Y|\mathbf{z}) = \boldsymbol{\theta}^\top (\mathbf{x} - \mathbb{E}(\mathbf{x}|\mathbf{z})) + \varepsilon, \quad (9)$$

which is a linear model between $Y - \mathbb{E}(Y|\mathbf{z})$ and $\mathbf{x} - \mathbb{E}(\mathbf{x}|\mathbf{z})$, and $\boldsymbol{\theta}$ can be directly estimated via ordinary least squares (OLS). After partialling out the nuisance, we still need to fit $\mathbb{E}(Y|\mathbf{z})$ and $\mathbb{E}(\mathbf{x}|\mathbf{z})$, but compared to g , these two models are tractable, at least, as we have both observations of input and output. However, some concerns remain. Firstly, we typically do not know any information about the parametric forms for $\mathbb{E}(Y|\mathbf{z})$ and $\mathbb{E}(\mathbf{x}|\mathbf{z})$, so it is more appropriate to estimate these conditional expectations using nonparametric or machine learning methods. Nevertheless, an important question is how to preserve the statistical efficiency of $\boldsymbol{\theta}$. Secondly, estimating $\mathbb{E}(Y|\mathbf{z})$, $\mathbb{E}(\mathbf{x}|\mathbf{z})$, and $\boldsymbol{\theta}$ using the same dataset may introduce bias into the estimation of $\boldsymbol{\theta}$. In fact, double machine learning (DML) alleviates the two issues by introducing the *Neyman orthogonality* condition and the *cross-fitting* technique.

2 Methodology

The key ideas underlying double machine learning are Neyman orthogonality and cross-fitting, which address potential bias from estimating nuisance functions. We first give the problem formulation generalized from Example 1.1, Example 1.2, and Example 1.3 in the previous section. In these examples, we are substantially dealing with the following moment condition with nuisance:

$$\mathbb{E}[\boldsymbol{\psi}(W; \boldsymbol{\theta}, \boldsymbol{\eta})] = \mathbf{0}, \quad (10)$$

where W denotes the observed data, $\boldsymbol{\theta}$ denotes the parameter of interest, $\boldsymbol{\eta}$ denotes the nuisance, and $\boldsymbol{\psi}$ denotes the score function (or moment function).

Suppose that we apply MoM or GMM to estimate $\boldsymbol{\theta}$ and $\boldsymbol{\eta}$ based on the moment condition (10). We denote $\boldsymbol{\theta}_0$ and $\boldsymbol{\eta}_0$ as the true parameter values, and $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\eta}}$ as the estimated values. **Our goal is to study the bias term $\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0$.** The true and estimated values satisfy:

$$P\boldsymbol{\psi}(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0) = \mathbf{0}, \quad (11)$$

$$P_n\boldsymbol{\psi}(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\eta}}) = \mathbf{0}, \quad (12)$$

where $Pf \triangleq \mathbb{E}(f(W; \cdot))$ and $P_nf \triangleq \frac{1}{n} \sum_{i=1}^n f(W_i; \cdot)$. We can derive the following decomposition equation from (11) and (12):

$$0 = \underbrace{(P_n - P)\boldsymbol{\psi}(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0)}_{\text{sampling noise}} + \underbrace{P\boldsymbol{\psi}(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\eta}})}_{\text{bias of } \boldsymbol{\psi}} + \underbrace{(P_n - P)\left[\boldsymbol{\psi}(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\eta}}) - \boldsymbol{\psi}(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0)\right]}_{\text{remainder}}. \quad (13)$$

For the bias term, we can further decompose it via Taylor expansion:

$$P\boldsymbol{\psi}(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\eta}}) = P\boldsymbol{\psi}(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0) + \partial_{\boldsymbol{\theta}} P\boldsymbol{\psi}(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0)(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + D_{\boldsymbol{\eta}} P\boldsymbol{\psi}(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0)[\hat{\boldsymbol{\eta}} - \boldsymbol{\eta}_0] + R_n, \quad (14)$$

where R_n denotes the high-degree remainder term. We define $J \triangleq \partial_{\boldsymbol{\theta}} \boldsymbol{\psi}(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0)$ and $\boldsymbol{\Psi} \triangleq P\boldsymbol{\psi}$ to simplify the notation, and we have

$$P\boldsymbol{\psi}(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\eta}}) = J(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + D_{\boldsymbol{\eta}} \boldsymbol{\Psi}(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0)[\hat{\boldsymbol{\eta}} - \boldsymbol{\eta}_0] + R_n. \quad (15)$$

Plugging (15) into (13) yields

$$\begin{aligned}
0 = & (P_n - P)\psi(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0) \\
& + J(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + D_\eta \Psi(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0)[\hat{\boldsymbol{\eta}} - \boldsymbol{\eta}_0] + R_n \\
& + (P_n - P) \left[\psi(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\eta}}) - \psi(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0) \right].
\end{aligned} \tag{16}$$

Therefore, we obtain

$$\begin{aligned}
\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0 = & - \underbrace{J^{-1}(P_n - P)\psi(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0)}_{\text{Term A}} \\
& - \underbrace{J^{-1}D_\eta \Psi(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0)[\hat{\boldsymbol{\eta}} - \boldsymbol{\eta}_0]}_{\text{Term B}} - \underbrace{J^{-1}(P_n - P) \left[\psi(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\eta}}) - \psi(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0) \right]}_{\text{Term C}} + R_n.
\end{aligned} \tag{17}$$

Thus, we decompose the error into three terms plus a high-order residual term. Term A arises from the sampling noise term in (13), Term B arises from the estimation error of $\boldsymbol{\eta}$, and Term C arises from the interaction between sampling noise and the estimation errors in $\boldsymbol{\theta}$ and $\boldsymbol{\eta}$. Although the sampling-noise term is unavoidable, Terms B and C can be effectively controlled using two key techniques in double machine learning: Neyman orthogonality and cross-fitting.

2.1 Neyman Orthogonality

After the derivation and analysis above, the motivation and requirement of Neyman orthogonality become straightforward. To eliminate Term B in (17), we would like to make the score function satisfy

$$D_\eta \Psi(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0) = \frac{d}{dr} \mathbb{E} [\psi(W, \boldsymbol{\theta}_0, \boldsymbol{\eta}_0 + r\mathbf{h})] = 0, \quad \forall \mathbf{h}, \tag{18}$$

which is defined by Gateaux derivative. Intuitively, the landscape of Ψ with respect to $\boldsymbol{\eta}$ at true parameter values is “flat”, so that order-one error in nuisance estimation will not matter.

We can verify that the score function constructed by the partialling-out technique in (9) is Neyman orthogonal. Let $m(\mathbf{z}) = \mathbb{E}(\mathbf{x}|\mathbf{z})$. The score function ψ and its expectation Ψ is

$$\psi(W; \boldsymbol{\theta}, g, m) = (\mathbf{x} - m(\mathbf{z})) (Y - \boldsymbol{\theta}^\top \mathbf{x} - g(\mathbf{z})), \tag{19}$$

$$\Psi(\boldsymbol{\theta}, g, m) = \mathbb{E} [\psi(W; \boldsymbol{\theta}, g, m)] = \mathbb{E} \{ (\mathbf{x} - m(\mathbf{z})) (Y - \boldsymbol{\theta}^\top \mathbf{x} - g(\mathbf{z})) \}. \tag{20}$$

For g , by letting $g_r = g_0 + rh_g$, we obtain

$$\Psi(\boldsymbol{\theta}_0, g_r, m_0) = \mathbb{E} \{ (\mathbf{x} - m_0(\mathbf{z})) (\varepsilon_Y - rh_g(\mathbf{z})) \}, \tag{21}$$

$$D_g \Psi[h_g] = -\mathbb{E} \{ (\mathbf{x} - m_0(\mathbf{z})) h_g(\mathbf{z}) \}. \tag{22}$$

Since $\mathbb{E} \{ (\mathbf{x} - m_0(\mathbf{z})) | \mathbf{z} \} = 0$, we have

$$D_g \Psi[h_g] = 0. \tag{23}$$

For m , by letting $m_r = m_0 + rh_m$, we obtain

$$\Psi(\boldsymbol{\theta}_0, g_0, m_r) = \mathbb{E} \{ (\mathbf{x} - m_0(\mathbf{z}) - rh_m(\mathbf{z})) \varepsilon_Y \}, \quad (24)$$

$$D_m \Psi[h_m] = -\mathbb{E} \{ h_m(\mathbf{z}) \varepsilon_Y \}. \quad (25)$$

Since

$$\mathbb{E}(\varepsilon_Y \mid \mathbf{z}) = 0, \quad (26)$$

the law of iterated expectations gives

$$\begin{aligned} D_m \Psi[h_m] &= -\mathbb{E} [h_m(\mathbf{z}) \mathbb{E}(\varepsilon_Y \mid \mathbf{z})] \\ &= 0. \end{aligned} \quad (27)$$

Therefore, the score is locally insensitive to first-order perturbations of both nuisance functions g and m at their true values. Hence,

$$D_g \Psi[h_g] = 0, \quad D_m \Psi[h_m] = 0, \quad (28)$$

for arbitrary admissible perturbation directions h_g and h_m , which establishes the Neyman orthogonality of the partialling-out score.

2.2 Cross-Fitting

Controlling Term C in (17) can be more difficult, as it involves both $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\eta}}$. To understand the main difficulty, first fix $\boldsymbol{\theta} = \boldsymbol{\theta}_0$ and consider

$$(P_n - P) [\psi(\boldsymbol{\theta}_0, \hat{\boldsymbol{\eta}}) - \psi(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0)]. \quad (29)$$

Define

$$\Delta_{\hat{\boldsymbol{\eta}}}(W) = \psi(W; \boldsymbol{\theta}_0, \hat{\boldsymbol{\eta}}) - \psi(W; \boldsymbol{\theta}_0, \boldsymbol{\eta}_0). \quad (30)$$

If $\hat{\boldsymbol{\eta}}$ were fixed and independent of the sample used to evaluate the empirical moment, then standard concentration arguments would give

$$(P_n - P)\Delta_{\hat{\boldsymbol{\eta}}} = O_p \left(\frac{\|\Delta_{\hat{\boldsymbol{\eta}}}\|_{P,2}}{\sqrt{n}} \right), \quad (31)$$

where

$$\|f\|_{P,2} = \{P[f(W)^2]\}^{1/2}. \quad (32)$$

Hence, if the nuisance estimator is consistent and the score is sufficiently regular such that

$$\|\Delta_{\hat{\boldsymbol{\eta}}}\|_{P,2} = o_p(1), \quad (33)$$

then

$$(P_n - P)\Delta_{\hat{\boldsymbol{\eta}}} = o_p(n^{-1/2}). \quad (34)$$

The difficulty is that, *without sample splitting*, $\hat{\boldsymbol{\eta}}$ is typically estimated using the same observations that are subsequently used to evaluate P_n . Therefore, $\Delta_{\hat{\boldsymbol{\eta}}}(W_i)$ is generally dependent on W_i , and the above conditional variance argument cannot be applied directly.

Controlling this dependence may require nontrivial empirical-process conditions on the class of nuisance estimators, such as Donsker or entropy conditions, which can be restrictive for flexible machine-learning methods.

Cross-fitting removes this dependence by separating nuisance estimation from score evaluation. Suppose the sample is partitioned into K disjoint folds,

$$I_1, \dots, I_K. \quad (35)$$

For each fold I_k , estimate the nuisance parameter using only observations outside that fold:

$$\hat{\boldsymbol{\eta}}^{(-k)} = \hat{\boldsymbol{\eta}}(\{W_i : i \notin I_k\}). \quad (36)$$

Then, for every $i \in I_k$, evaluate the score using

$$\psi(W_i; \boldsymbol{\theta}, \hat{\boldsymbol{\eta}}^{(-k)}). \quad (37)$$

The resulting cross-fitted empirical moment is

$$\frac{1}{n} \sum_{k=1}^K \sum_{i \in I_k} \psi(W_i; \boldsymbol{\theta}, \hat{\boldsymbol{\eta}}^{(-k)}). \quad (38)$$

Conditional on the training sample used to construct $\hat{\boldsymbol{\eta}}^{(-k)}$, this nuisance estimator can be treated as fixed, while the observations in I_k remain independent of it. Therefore, for

$$\Delta_k(W) = \psi(W; \boldsymbol{\theta}_0, \hat{\boldsymbol{\eta}}^{(-k)}) - \psi(W; \boldsymbol{\theta}_0, \boldsymbol{\eta}_0), \quad (39)$$

we have

$$\mathbb{E}[(P_{n,k} - P)\Delta_k \mid \hat{\boldsymbol{\eta}}^{(-k)}] = 0, \quad (40)$$

and

$$\text{Var}[(P_{n,k} - P)\Delta_k \mid \hat{\boldsymbol{\eta}}^{(-k)}] \leq \frac{\|\Delta_k\|_{P,2}^2}{|I_k|}, \quad (41)$$

where $P_{n,k}$ denotes the empirical measure over fold I_k . Consequently,

$$(P_{n,k} - P)\Delta_k = O_p\left(\frac{\|\Delta_k\|_{P,2}}{\sqrt{|I_k|}}\right). \quad (42)$$

If

$$\|\Delta_k\|_{P,2} = o_p(1), \quad (43)$$

then the nuisance-induced empirical-process term is

$$o_p(n^{-1/2}). \quad (44)$$

Thus, the main role of cross-fitting is not to reduce the population bias caused by nuisance estimation—this is the role of Neyman orthogonality—but rather to control the empirical

dependence between the estimated nuisance parameter and the observations used to evaluate the score.

Finally, Term C also contains the effect of replacing $\boldsymbol{\theta}_0$ by $\hat{\boldsymbol{\theta}}$. It can be decomposed as

$$\begin{aligned} & (P_n - P) \left[\psi(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\eta}}) - \psi(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0) \right] \\ &= (P_n - P) \left[\psi(\boldsymbol{\theta}_0, \hat{\boldsymbol{\eta}}) - \psi(\boldsymbol{\theta}_0, \boldsymbol{\eta}_0) \right] \\ & \quad + (P_n - P) \left[\psi(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\eta}}) - \psi(\boldsymbol{\theta}_0, \hat{\boldsymbol{\eta}}) \right]. \end{aligned} \tag{45}$$

Cross-fitting mainly controls the first term. The second term is typically handled using the consistency of $\hat{\boldsymbol{\theta}}$, together with smoothness and stochastic equicontinuity of the score. Under suitable regularity conditions, both terms are $o_p(n^{-1/2})$, so Term C does not contribute to the first-order asymptotic distribution.